

Sheth L.U.J. College And Sir M.V. College of Science

#1.A Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques

#CLI Program

```
def encrypt(text, key):
    result = ""
    for ch in text:
        if ch.isalpha():
            base = ord('A') if ch.isupper() else ord('a')
            result += chr((ord(ch)-base+key)%26+base)
        else:
            result += ch
    return result
```

```
def decrypt(text, key):
    result = ""
    for ch in text:
        if ch.isalpha():
            base = ord('A') if ch.isupper() else ord('a')
            result += chr((ord(ch)-base-key)%26+base)
        else:
            result += ch
    return result
```

```
text = input("Enter Message: ")
key = int(input("Enter Key: "))
```

```
cipher = encrypt(text, key)
print("Encrypted:", cipher)
```

```
plain = decrypt(cipher, key)
print("Decrypted:", plain)
```

#GUI Program
import tkinter as tk

```
def encrypt():
    text = entry.get()
    key = int(key_entry.get())
    result = ""
    for ch in text:
```

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Sheth L.U.J. College And Sir M.V. College of Science

```
        if ch.isalpha():
            base = ord('A') if ch.isupper() else ord('a')
            result += chr((ord(ch)-base+key)%26+base)
        else:
            result += ch
    output.config(text="Encrypted: " + result)

def decrypt():
    text = entry.get()
    key = int(key_entry.get())
    result = ""
    for ch in text:
        if ch.isalpha():
            base = ord('A') if ch.isupper() else ord('a')
            result += chr((ord(ch)-base-key)%26+base)
        else:
            result += ch
    output.config(text="Decrypted: " + result)

root = tk.Tk()
root.title("Caesar Cipher")

tk.Label(root, text="Message").pack()
entry = tk.Entry(root, width=40)
entry.pack()

tk.Label(root, text="Key").pack()
key_entry = tk.Entry(root)
key_entry.pack()

tk.Button(root, text="Encrypt", command=encrypt).pack(pady=5)
tk.Button(root, text="Decrypt", command=decrypt).pack(pady=5)

output = tk.Label(root, text="")
output.pack()

root.mainloop()
```

#1. B Design and implement algorithms to encrypt and decrypt messages using transposition techniques

```
def encrypt(text, key):
```

```
    result = ""
```

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Roll no:- TY106

Sub:- cyber security

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```
for ch in text:
    if ch.isalpha():
        base = ord('A') if ch.isupper() else ord('a')
        result += chr((ord(ch) - base + key) % 26 + base)
    else:
        result += ch
return result
```

```
def decrypt(text, key):
    result = ""
    for ch in text:
        if ch.isalpha():
            base = ord('A') if ch.isupper() else ord('a')
            result += chr((ord(ch) - base - key) % 26 + base)
        else:
            result += ch
    return result
```

```
text = input("Enter message: ")
key = int(input("Enter key: "))
```

```
cipher = encrypt(text, key)
print("Encrypted:", cipher)
```

```
plain = decrypt(cipher, key)
print("Decrypted:", plain)
```

```
#GUI Program (Tkinter)
import tkinter as tk
```

```
def encrypt():
    text = msg.get()
    key = int(rail.get())

    rails = [" " for _ in range(key)]
    row = 0
    direction = 1

    for ch in text:
        rails[row] += ch
        row += direction

        if row == 0 or row == key-1:
            direction *= -1
```

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Roll no:- TY106
Sub:- cyber security

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```
output.config(text="Encrypted: " + ".join(rails))

root = tk.Tk()
root.title("Rail Fence Cipher")

tk.Label(root, text="Message").pack()
msg = tk.Entry(root, width=40)
msg.pack()

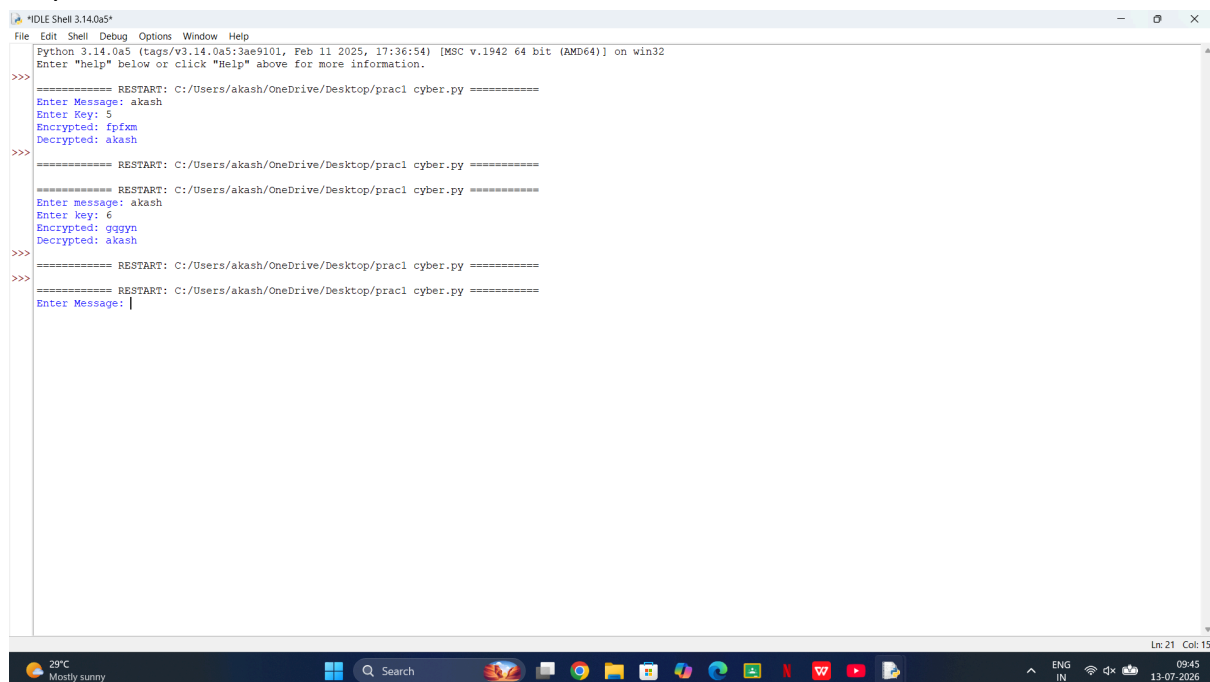
tk.Label(root, text="Rails").pack()
rail = tk.Entry(root)
rail.pack()

tk.Button(root, text="Encrypt", command=encrypt).pack()

output = tk.Label(root, text="")
output.pack()

root.mainloop()
```

Output:-



```
*IDLE Shell 3.14.0a5*
File Edit Shell Debug Options Window Help
Python 3.14.0a5 (tags/v3.14.0a5:3ae9101, Feb 11 2025, 17:36:54) [MSC v.1942 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/akash/OneDrive/Desktop/prac1 cyber.py =====
Enter Message: akash
Enter Key: 5
Encrypted: fpfxm
Decrypted: akash
>>>
===== RESTART: C:/Users/akash/OneDrive/Desktop/prac1 cyber.py =====
===== RESTART: C:/Users/akash/OneDrive/Desktop/prac1 cyber.py =====
Enter message: akash
Enter key: 6
Encrypted: gqgyn
Decrypted: akash
>>>
===== RESTART: C:/Users/akash/OneDrive/Desktop/prac1 cyber.py =====
>>>
===== RESTART: C:/Users/akash/OneDrive/Desktop/prac1 cyber.py =====
Enter Message: |
```

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Message

Akash

Key

5

Encrypt

Decrypt

Encrypted: Fpfxm

Message

Akash

Key

5

Encrypt

Decrypt

Decrypted: Vfvnc

Message

Akash

Rails

4

Encrypt

Encrypted: Akahs

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